

Lecture - 37

Laplace Transform

Chapter 9 of Oppenheim, Willsky and Nawab

Recap:

① $e^{-at} u(t) \xleftrightarrow{L} \frac{1}{s+a}, \operatorname{Re}(s) > -\operatorname{Re}(a)$

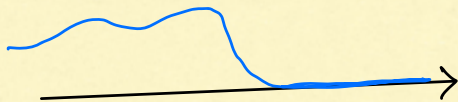
② $-e^{-at} u(-t) \xleftrightarrow{L} \frac{1}{s+a}, \operatorname{Re}(s) < -\operatorname{Re}(a)$

③ Rational $X(s) = \frac{N(s)}{D(s)}$

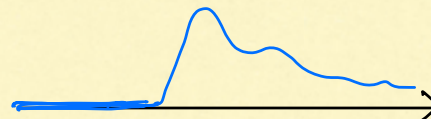
- Poles: roots of $D(s)$
- Zeros: roots of $N(s)$

④ "Sidedness":

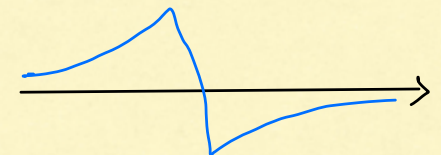
Left-sided



Right-sided



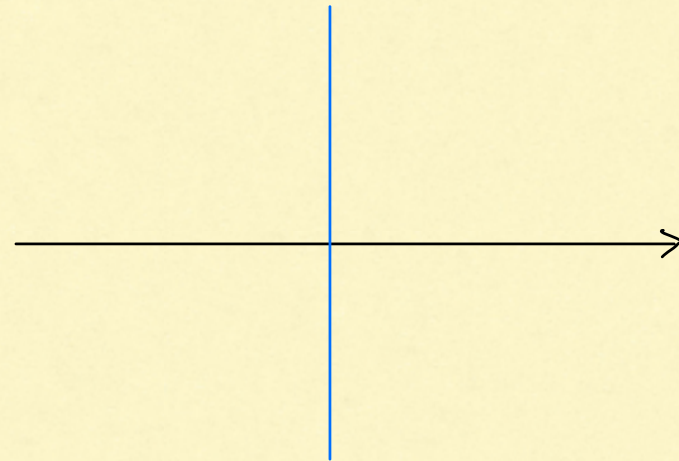
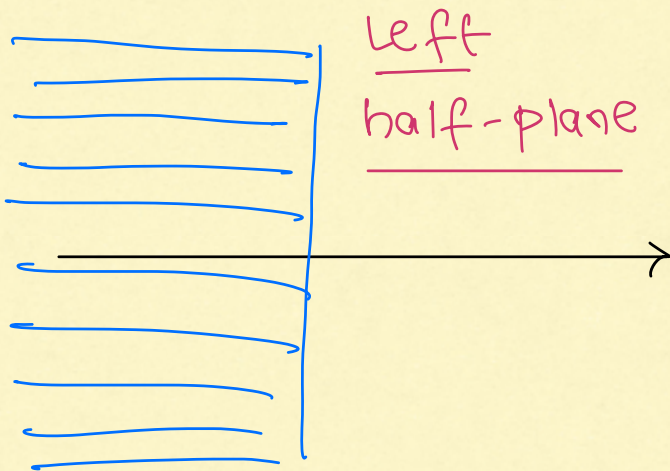
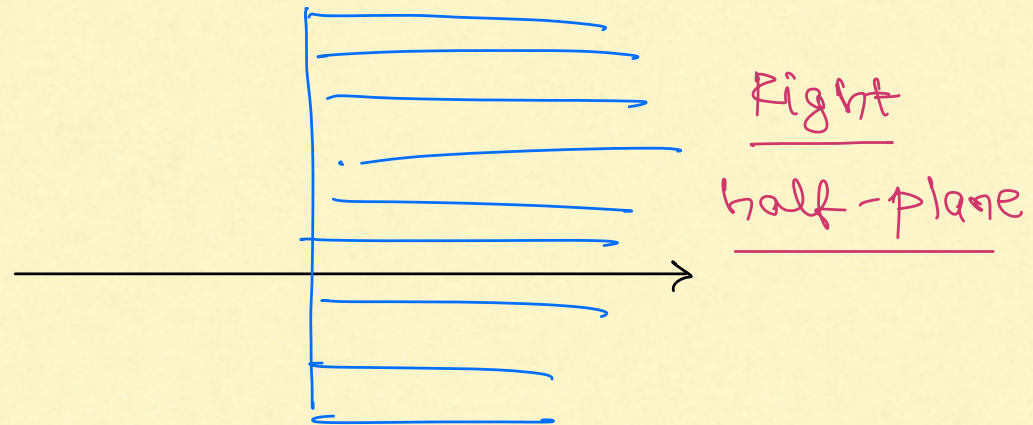
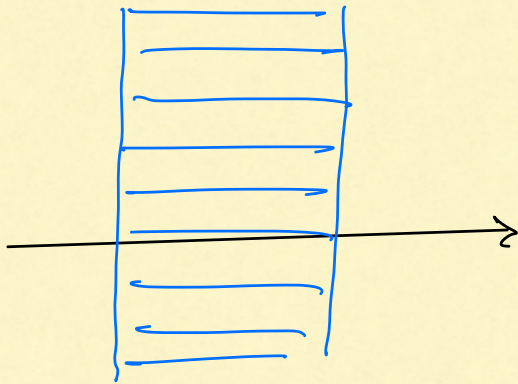
Two-sided



I

Properties of ROC

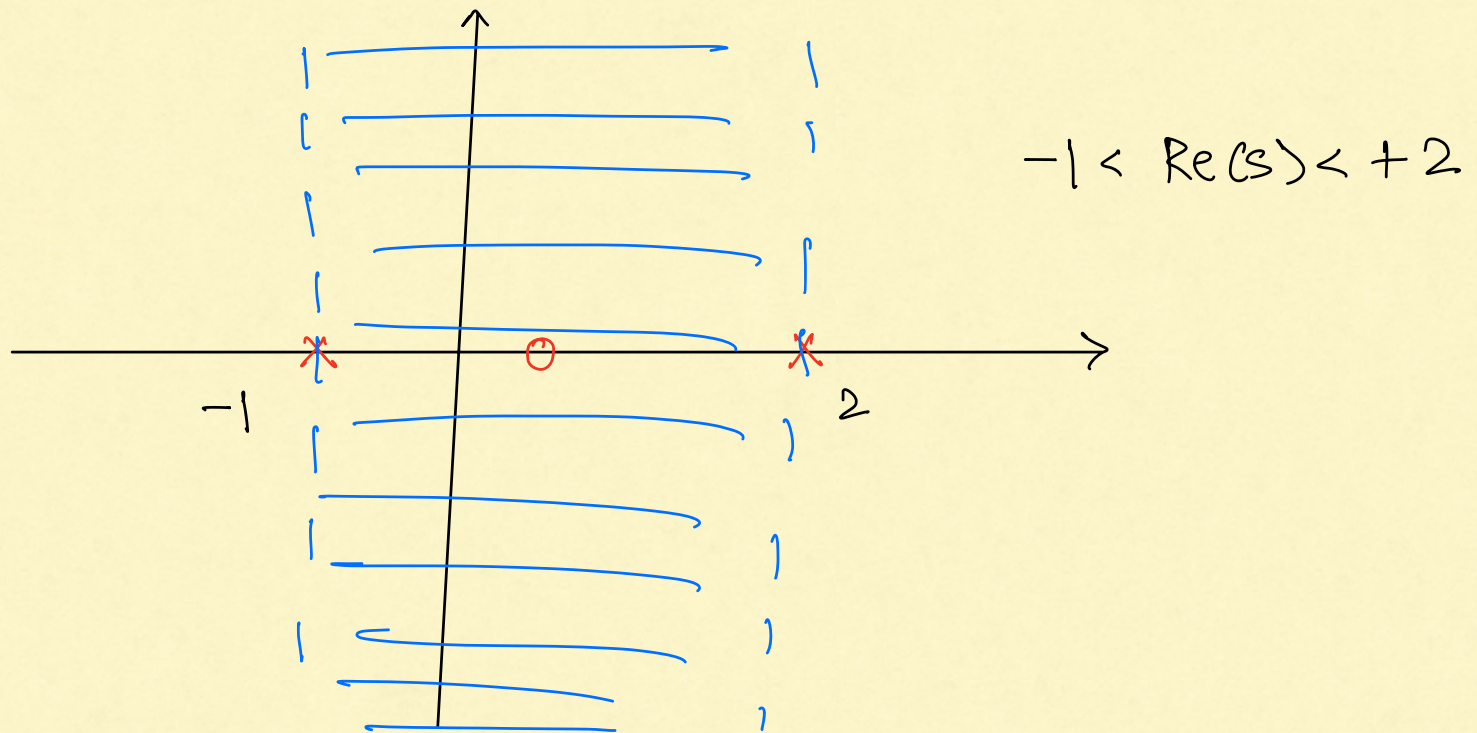
- ① ROC is a strip in the s -plane parallel to $j\omega$ -axis.



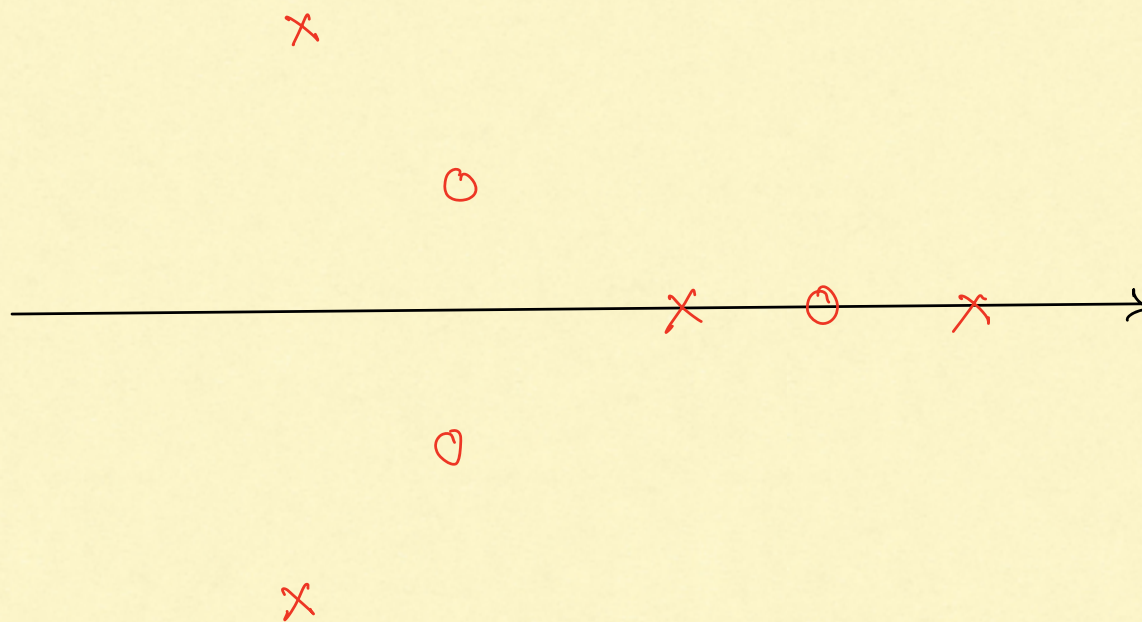
ROC may/may not include the boundary line.

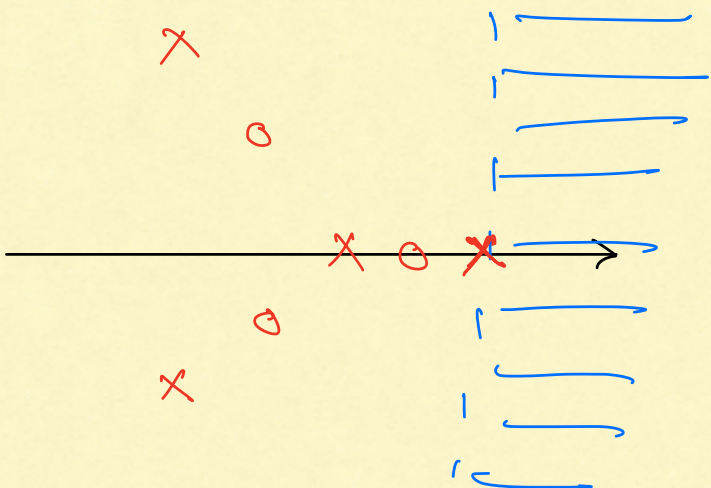
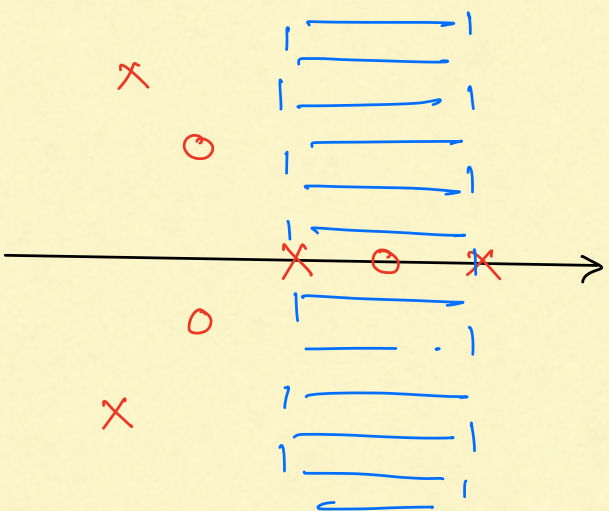
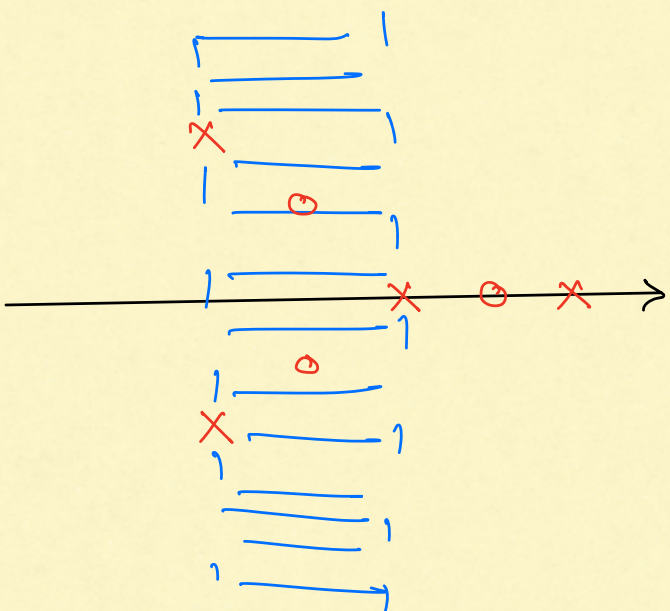
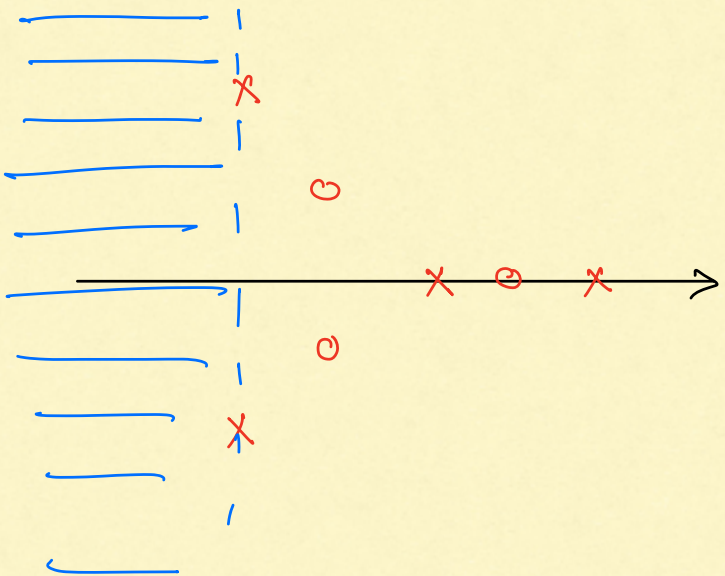
② If $X(s)$ is rational:

- ROC does not contain any poles.
- ROC is bounded by poles or extends to infinity



ROC may or may not contain zeroes.





3

x is finite-duration

and

$$\int_{-\infty}^{+\infty} |x(t)| dt < \infty$$

$\Rightarrow$

ROC is entire s -plane

x is left-sided $\Rightarrow$

ROC is entire s -plane or

ROC is a left half-plane

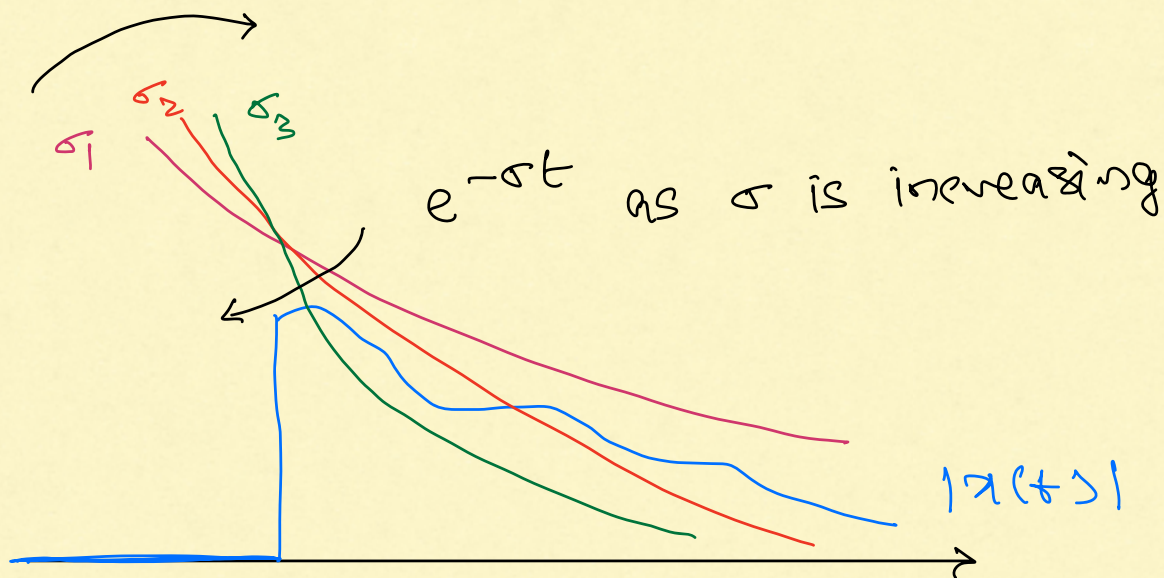
x is right-sided $\Rightarrow$

ROC is entire s -plane or

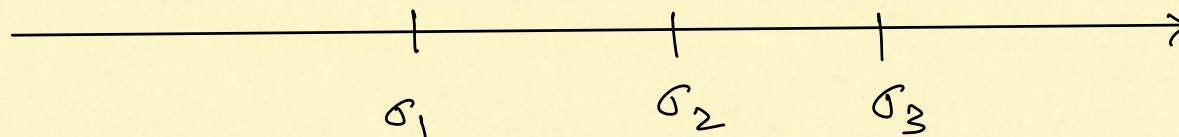
ROC is a right half-plane

(x is two-sided : anything is possible)

σ increasing



Right-sided
Signal



If $\text{Re}(s) = \sigma_1$ is in the ROC, then
 $\text{Re}(s) > \sigma_1$ is in the ROC too.

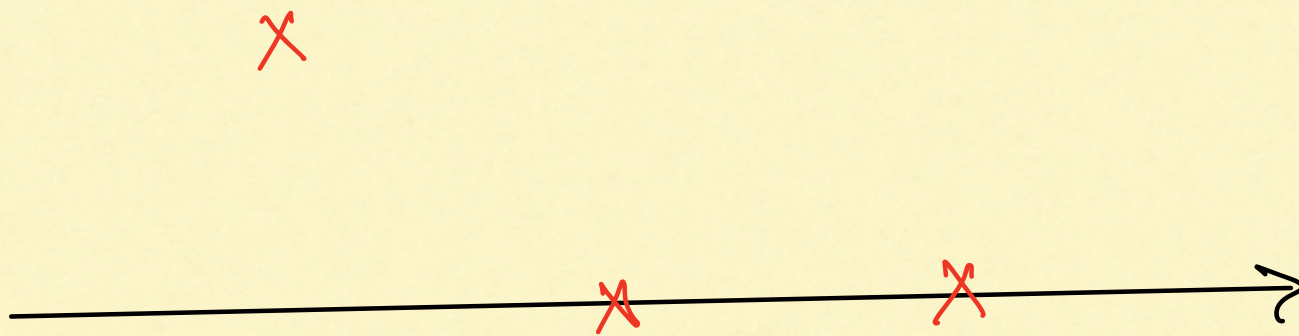
④

If $X(s)$ is rational:

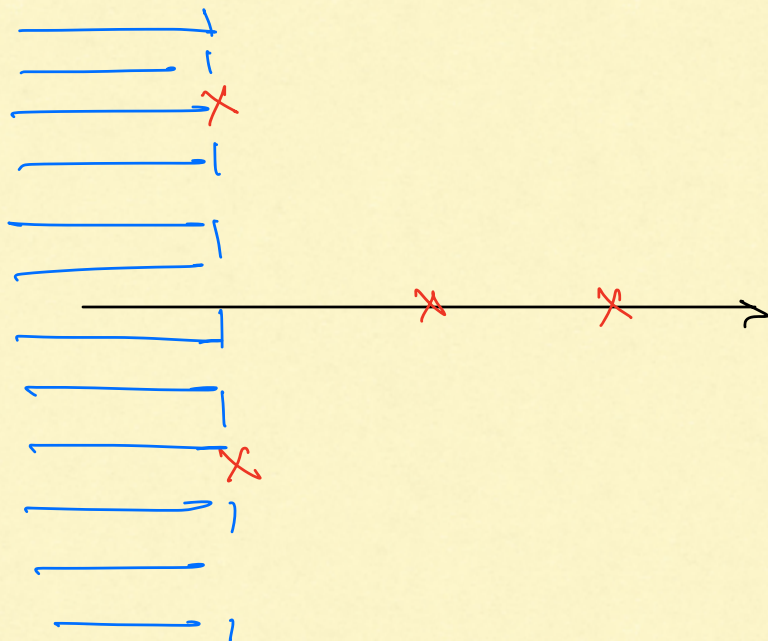
x is right-sided $\Rightarrow$ ROC is the right half plane to the right of right most pole.

x is left-sided $\Rightarrow$ ROC is the left half-plane to the left of the left most pole.

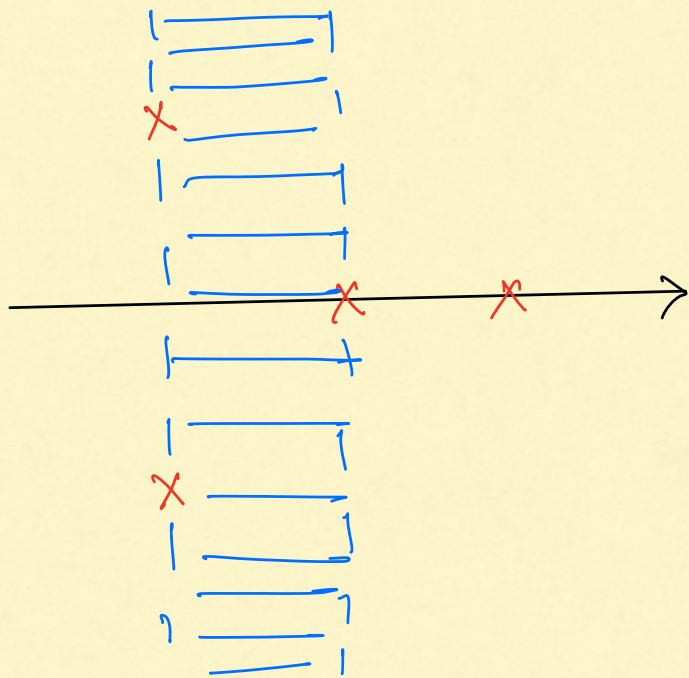
x is two-sided $\Rightarrow$ Bounded by poles on left and right.



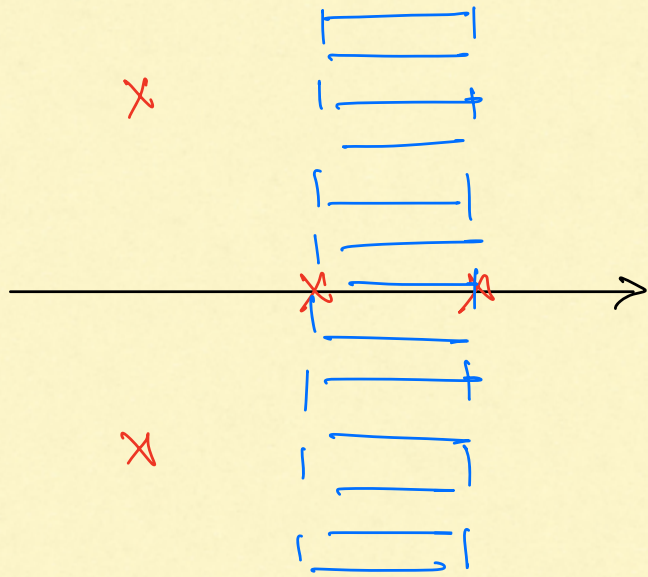
.



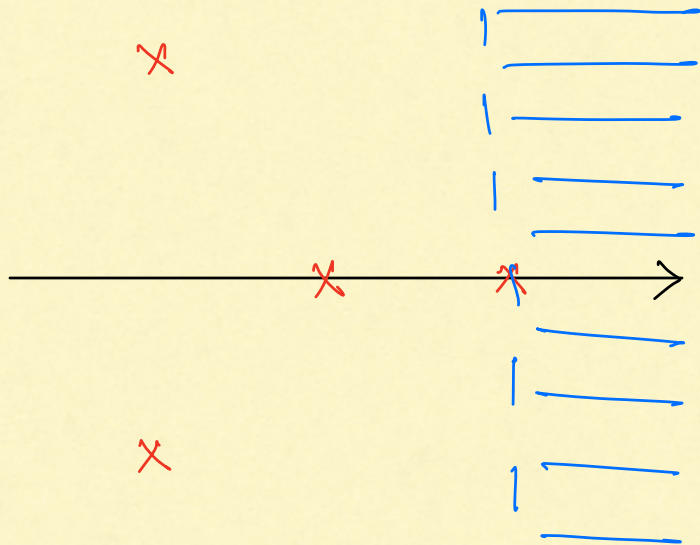
$x(t)$ is
left-sided.



$x(t)$ is two-sided



$x(t)$ is two-sided.



$x(t)$ is right-sided

II

Properties of Laplace Transform

①

Linearity

$$\text{If } x_1 \xleftrightarrow{L} X_1, \text{ ROC } R_1$$

$$x_2 \xleftrightarrow{L} X_2, \text{ ROC } R_2$$

then

$$\alpha_1 x_1 + \alpha_2 x_2 \xleftrightarrow{L} \alpha_1 X_1 + \alpha_2 X_2,$$

ROC includes $R_1 \cap R_2$

② Time shifting

$$x(t) \xleftrightarrow{L} X(s), \quad \text{ROC } R$$

$$\Rightarrow x(t - t_0) \xleftrightarrow{L} e^{-st_0} X(s), \quad \text{ROC } R.$$

Ex: $e^{-at} u(t) \xleftrightarrow{L} \frac{1}{s+a}, \quad \text{Re}(s) > -\text{Re}(a).$

$$e^{-a(t-t_0)} u(t-t_0) \xleftrightarrow{L} \frac{e^{-st_0}}{s+a}, \quad \text{Re}(s) > -\text{Re}(a)$$

③ shifting in s-Domain:

$$x(t) \xleftrightarrow{L} X(s), \text{ ROC: } s \in \mathbb{R}$$

$$\Rightarrow e^{s_0 t} x(t) \xleftrightarrow{L} X(s - s_0), \text{ ROC: } s - s_0 \in \mathbb{R}$$

④ Time scaling

$$x(t) \xleftrightarrow{L} X(s), \quad \text{ROC: } s \in \mathbb{R}$$

$a \in \mathbb{R}, a \neq 0$:

$$x(at) \xleftrightarrow{L} \frac{1}{|a|} X\left(\frac{s}{a}\right), \quad \text{ROC: } \frac{s}{a} \in \mathbb{R}$$

$$x(-t) \xleftrightarrow{L} X(-s), \quad \text{ROC: } -s \in \mathbb{R}$$

⑤ conjugation

$$x(t) \xleftrightarrow{L} X(s), \quad \text{ROC: } s \in R$$

$$x^*(t) \xleftrightarrow{L} X^*(s^*), \quad \text{ROC: } s \in R$$

$$X^*(s^*) = [X(s^*)]^*$$

6

Convolution:

$$x_1(t) * x_2(t) \xleftrightarrow{L} X_1(s) \cdot X_2(s),$$

ROC contains $R_1 \cap R_2$

⑦ and ⑧ Derivative Theorems

$$x(t) \xleftrightarrow{L} X(s), \quad \text{ROC: } s \in R$$

$$\frac{dx}{dt} \xleftrightarrow{L} s X(s), \quad \text{ROC: contains } R$$

$$-t x(t) \xleftrightarrow{L} \frac{dX}{ds}, \quad \text{ROC} = R$$

Example:

$$e^{-at} u(t) \xleftrightarrow{L} \frac{1}{s+a}, \quad \operatorname{Re}(s) > -\operatorname{Re}(a)$$

$$-t \cdot e^{-at} u(t) \xleftrightarrow{L}$$

$$\therefore t \cdot e^{-at} u(t) \xleftrightarrow{L}$$

⑨ Integration

$$x(t) \xleftrightarrow{L} X(s), \quad \text{ROC} = \mathbb{R}$$

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{L} \frac{1}{s} X(s), \quad \text{ROC contains} \\ \mathbb{R} \cap \{s : \operatorname{Re}(s) > 0\}$$

||

$$x(t) * u(t)$$

10

Initial & Final Value Theorems

Assuming: x is a function (not a distribution)

$$\& x(t) = 0 \quad \forall t < 0.$$

$$\bullet \quad x(0^+) = \lim_{s \rightarrow \infty} s X(s)$$

$$\bullet \quad \lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} s X(s)$$

III

Laplace Transform Pairs

<u>Signal</u>	<u>Transform</u>	<u>Roc</u>
$\delta(t)$	1	$\mathbb{C}$
$\delta^{(n)}(t)$	s^n	$\mathbb{C}$
$u(t)$	$1/s$	$\text{Re}(s) > 0$
$-u(-t)$	$1/s$	$\text{Re}(s) < 0$
$\frac{t^{n-1}}{(n-1)!} u(t)$	$1/s^n$	$\text{Re}(s) > 0$
$= u(t) * \dots * u(t)$		

$$e^{-\alpha t} \cos \omega_0 t u(t) \\ (\alpha \in \mathbb{R})$$

$$\frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2}$$

$$\operatorname{Re}(s) > -\operatorname{Re}(\alpha)$$

$$e^{-\alpha t} \sin \omega_0 t u(t) \\ (\alpha \in \mathbb{R})$$

$$\frac{\omega_0}{(s + \alpha)^2 + \omega_0^2}$$

$$\operatorname{Re}(s) > -\operatorname{Re}(\alpha)$$

$$\frac{t^{n-1}}{(n-1)!} e^{-\alpha t} u(t)$$

$$\frac{1}{(s + \alpha)^n}$$

$$\operatorname{Re}(s) > -\operatorname{Re}(\alpha)$$

$$-\frac{t^{n-1}}{(n-1)!} e^{-\alpha t} u(-t)$$

$$\frac{1}{(s + \alpha)^n}$$

$$\operatorname{Re}(s) < -\operatorname{Re}(\alpha)$$

IV

Inverting a Rational Laplace Transform

1

$$X(s) = \frac{N(s)}{D(s)}$$

long division

$$= \underline{f(s)} +$$

$$\frac{r(s)}{D(s)}$$

} degree of r
<
degree of D

(Cancel common factors)

Polynomial

Partial fraction expansion

Partial fraction expansion

$$= f(s) + \sum_i \frac{A_i \rightarrow \text{constants}}{s + a_i \rightarrow \text{Poles } -a_i}$$

Example:

$$X(s) = \frac{2s^2 + 5s + 4}{(s+1)(s+2)}$$

$$= \frac{2s^2 + 5s + 4}{s^2 + 3s + 2}$$

$$\begin{array}{r} s^2 + 3s + 2 \overline{) 2s^2 + 5s + 4} \\ \underline{2s^2 + 6s + 4} \\ -s + 0 \\ \hline \end{array}$$

$$X(s) = 2 - \frac{s}{(s+1)(s+2)}$$

$$\frac{s}{(s+1)(s+2)} = \frac{A_1}{s+1} + \frac{A_2}{s+2}$$

$$= \frac{(A_1 + A_2)s + (2A_1 + A_2)}{(s+1)(s+2)}$$

$$\Rightarrow 2A_1 + A_2 = 0$$

$$A_1 + A_2 = 1$$

$$A_1 = -1$$

$$A_2 = 2$$

$$\therefore \frac{s}{(s+1)(s+2)} = \frac{-1}{s+1} + \frac{2}{s+2}$$

$$\therefore X(s) = 2 + \frac{1}{s+1} - \frac{2}{s+2}$$

② Invert term-by-term:

$$X(s) = f(s) + \sum_i \frac{A_i}{s + a_i}$$

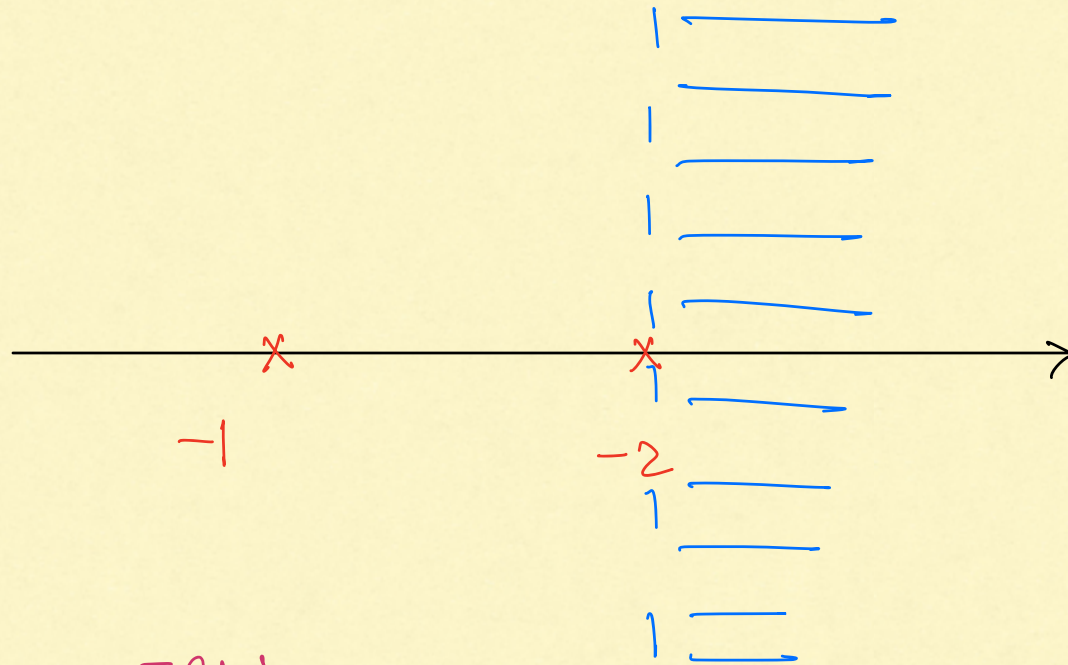
$$= b_0 + b_1 s + \dots + b_n s^n + \sum_i \frac{A_i}{s + a_i}$$

$$b_k s^k \rightarrow b_k \cdot \delta^{(k)}(t)$$

$$\frac{A_i}{s + a_i} \rightarrow \begin{cases} A_i \cdot e^{-a_i t} u(t) & \text{if ROC is on the right of } -a_i \\ -A_i e^{-a_i t} u(-t) & \text{if ROC is on the left of } -a_i \end{cases}$$

Example: $X(s) = 2 + \frac{1}{s+1} - \frac{2}{s+2}$

RCC:



$2 \rightarrow 2\delta(t)$

$\frac{1}{s+1} \rightarrow e^{-t}u(t)$

$\frac{2}{s+2} \rightarrow 2 \cdot e^{-2t}u(t)$

$\therefore x(t) = 2\delta(t)$

$+ e^{-t}u(t)$

$- 2 \cdot e^{-2t}u(t)$